

Multi-Pair Electron–Positron Production in Muon Decay: Experimental Frontier from Diagram Enumeration and NDA

Tony Menzo and Claude Opus 4.6 (high effort)

Computed with HEPTAPOD

March 19, 2026

Abstract

We systematically study the Standard Model decay $\mu^+ \rightarrow \bar{\nu}_\mu \nu_e e^+ + n(e^+e^-)$ for $n = 0, 1, 2, 3$ electron–positron pairs. Using automated tree-level diagram enumeration, naive dimensional analysis (NDA), and exact MadGraph calculations, we determine the branching ratio as a function of the pair multiplicity n . The number of contributing Feynman diagrams grows from 1 at $n = 0$ to 149,400 at $n = 3$, yet the rate is dominated at every multiplicity by diagrams with a single heavy W propagator. MadGraph cross-checks at $n = 1$ and $n = 2$ validate the NDA estimates to within a factor of a few, confirming $\text{BR}(n=1) = 3.56 \times 10^{-5}$ and $\text{BR}(n=2) = 4.34 \times 10^{-10}$. The effective per-pair suppression factor is approximately $(\alpha/\pi)^2 \approx 5 \times 10^{-6}$, modulated by a rapidly growing diagram multiplicity that partially compensates the phase-space penalty. We identify $n = 3$ ($\text{BR} \sim 10^{-13}$) as the experimental frontier for Mu3e Phase I, and $n = 4$ ($\text{BR} \sim 10^{-16}$) as marginally accessible at Mu3e Phase II.

1 Introduction

The dominant decay of the muon, $\mu^+ \rightarrow e^+ \nu_e \bar{\nu}_\mu$, proceeds through a single W boson exchange with unit branching ratio. Higher-multiplicity variants in which one or more virtual photons radiated from the charged lepton lines convert into e^+e^- pairs provide a clean probe of higher-order electroweak dynamics. The $n = 1$ channel, $\mu^+ \rightarrow e^+e^+e^- \nu_e \bar{\nu}_\mu$, was measured by SINDRUM [1] at $\text{BR} = (3.4 \pm 0.4) \times 10^{-5}$, and NLO corrections have been computed [2, 3]. The $n = 2$ channel (seven-body) has been predicted at $\text{BR} = (3.93 \pm 0.01) \times 10^{-10}$ by Hostert *et al.* [4] and is expected to be first observed at Mu3e Phase I with $\sim 1,000$ events. No prediction exists for $n \geq 3$.

The question we address is: *what is the largest n for which $\mu^+ \rightarrow \bar{\nu}_\mu \nu_e e^+ + n(e^+e^-)$ remains experimentally observable?* The answer is determined by the interplay between the per-pair suppression (coupling, phase space, propagator) and the combinatorial explosion of contributing Feynman diagrams.

Experimental landscape. The relevant current and planned muon experiments are:

- **SINDRUM (1985):** Measured $\text{BR}(n=1)$ from 7.3×10^{12} stopped muons. Sensitivity $\sim 10^{-5}$.
- **Mu3e Phase I** (PSI, 2025–2026): $10^8 \mu^+/\text{s}$ on the πE5 beamline, 2.5×10^{15} total stops, single-event sensitivity (SES) $= 2 \times 10^{-15}$.
- **Mu3e Phase II** (PSI, ~ 2030): $10^9 \mu^+/\text{s}$ with the HiMB upgrade, 5.5×10^{16} total stops, $\text{SES} \sim 10^{-16}$.
- **HiMB future:** Up to $10^{10} \mu^+/\text{s}$, pushing the BR floor toward $\sim 10^{-18}$.

2 Method

Our analysis pipeline combines three complementary tools:

Diagram enumeration. FEYNGRAPH enumerates all tree-level Feynman diagrams for a given process and classifies them by the number of heavy (W , Z) propagators. Each additional heavy propagator introduces a suppression $(m_\mu/M_W)^4 \sim 10^{-13}$, establishing a clear hierarchy among diagram classes.

Naive dimensional analysis (NDA). For each diagram class, NDA estimates the partial width from the coupling structure, propagator masses, and n -body phase-space volume Φ_n . The massless n -body phase space scales as

$$\Phi_n = \frac{M^{2n-4}}{2^{4n-5} \pi^{2n-3} (n-1)! (n-2)!}, \quad (1)$$

where $M = m_\mu$ is the mother particle mass.

MadGraph. Exact tree-level widths are computed with MadGraph5_aMC@NLO using the `sm-lepton_masses` model (which retains $m_\mu, m_e \neq 0$). Coupling-order restrictions ($\text{QED} \leq 4$, $\text{QED} \leq 6$) isolate the dominant diagram class for independent cross-checking.

3 Results

3.1 Diagram enumeration

Table 1 summarizes the diagram enumeration for each multiplicity. The total number of Feynman diagrams grows super-exponentially: $1 \rightarrow 18 \rightarrow 1,122 \rightarrow 149,400$ from $n = 0$ to $n = 3$, driven by the combinatorial ways of attaching photon propagators to the increasing number of charged lines.

Table 1: Tree-level Feynman diagram count by heavy-propagator class for $\mu^+ \rightarrow \bar{\nu}_\mu \nu_e e^+ + n(e^+e^-)$. The dominant class (fewest heavy propagators) is highlighted.

n	Particles	Diagrams by heavy-propagator count							Total
		1W	2W	3W	4W	5W	6W	7W	
0	3	1	–	–	–	–	–	–	1
1	5	4	12	2	–	–	–	–	18
2	7	84	378	504	138	18	–	–	1,122
3	9	4,320	24,768	52,272	47,520	16,416	3,744	360	149,400

The dominant-class fraction shrinks rapidly: 100% at $n = 0$, 22% at $n = 1$, 7.5% at $n = 2$, and 2.9% at $n = 3$. Yet the subleading classes are suppressed by $(m_\mu/M_W)^4 \sim 10^{-13}$ per additional heavy propagator in the rate, so they contribute negligibly to the total width at every n .

Figure 1 shows representative dominant-class diagrams. At each multiplicity, the dominant topology consists of a single W exchange (the Michel vertex) dressed by n virtual photons that each convert to an e^+e^- pair. The couplings for the dominant class are $g_w^2 e^{2n}$, confirming the purely electromagnetic character of the additional pair production.

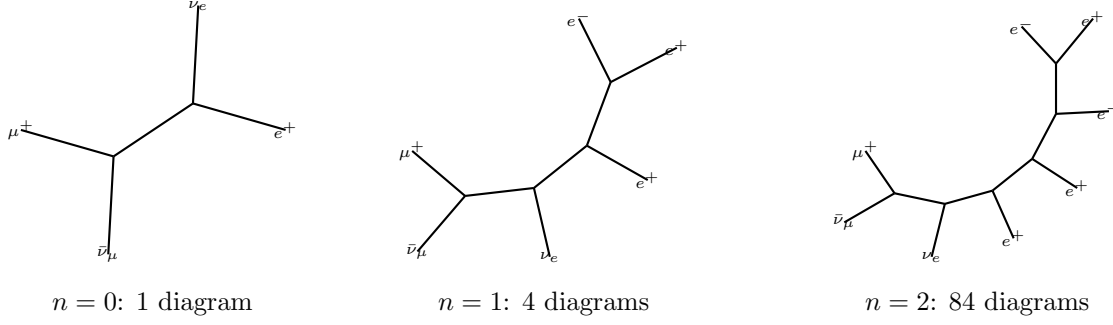


Figure 1: Representative dominant-class (single- W) Feynman diagrams for $n = 0, 1, 2$ electron-positron pairs. Each additional pair arises from a virtual photon (wavy line) radiated off an existing charged-fermion line. The dominant-class diagram count and coupling structure ($g_w^2 e^{2n}$) are indicated.

3.2 NDA branching ratio estimates

Table 2 presents the NDA branching ratio estimates for each multiplicity and diagram class. In every case, the single- W class overwhelmingly dominates, with subleading classes suppressed by at least 10^{13} per additional heavy propagator.

Table 2: NDA partial widths and branching ratios for the dominant ($1W$) class at each pair multiplicity. The NDA formula, number of diagrams, width per diagram, total class width, and branching ratio are shown. The reference total width is $\Gamma_\mu = 2.996 \times 10^{-19}$ GeV.

n	Diagrams	NDA formula	Width/diag (GeV)	Total width (GeV)	BR
0	1	$\frac{g_w^4 M^5}{64\pi^3 M_W^4}$	8.56×10^{-19}	8.56×10^{-19}	2.86
1	4	$\frac{g_w^4 e^4 M^5}{\pi^7 M_W^4}$	1.39×10^{-24}	5.55×10^{-24}	1.85×10^{-5}
2	84	$\frac{g_w^4 e^8 M^5}{\pi^{11} M_W^4}$	6.73×10^{-30}	5.65×10^{-28}	1.89×10^{-9}
3	4,320	$\frac{g_w^4 e^{12} M^5}{\pi^{15} M_W^4}$	5.73×10^{-35}	2.47×10^{-31}	8.26×10^{-13}

The NDA estimate at $n = 0$ gives $\text{BR} = 2.86$, an $O(1)$ number consistent with the known result (the Michel decay saturates the total width). The NDA accuracy is typical: correct to within a factor of a few at each multiplicity, as confirmed by MadGraph in the following section.

3.3 MadGraph cross-checks

Table 3 compares NDA and MadGraph results for $n = 1$ and $n = 2$. In both cases, restricting MadGraph to the dominant coupling order ($\text{QED} \leq 4$ for $n = 1$, $\text{QED} \leq 6$ for $n = 2$) reproduces the full result, confirming that subleading diagram classes are negligible.

Key observations:

- At $n = 1$, the MadGraph result $\text{BR} = 3.56 \times 10^{-5}$ agrees with the SINDRUM measurement $(3.4 \pm 0.4) \times 10^{-5}$ to within one sigma.
- At $n = 2$, MadGraph gives $\text{BR} = 4.34 \times 10^{-10}$, consistent with the prediction of $(3.93 \pm 0.01) \times 10^{-10}$ from Ref. [4].

Table 3: Comparison of NDA estimates with exact MadGraph tree-level widths for the dominant diagram class.

n	NDA width (GeV)	MG width (GeV)	NDA BR	MG BR	Ratio
1	5.55×10^{-24}	1.066×10^{-23}	1.85×10^{-5}	3.56×10^{-5}	0.52
2	5.65×10^{-28}	1.301×10^{-28}	1.89×10^{-9}	4.34×10^{-10}	4.3

- The NDA-to-MadGraph ratio varies from 0.52 to 4.3, typical of NDA accuracy. At $n = 1$, NDA underestimates by a factor of 2; at $n = 2$, it overestimates by a factor of 4. This variation reflects the growing importance of destructive interference among diagrams at higher multiplicity, which NDA does not capture.

4 Per-pair suppression factor

The effective branching ratio suppression per additional e^+e^- pair results from the competition of three factors:

Coupling. Each pair introduces two QED vertices (photon emission and $\gamma \rightarrow e^+e^-$ conversion), contributing a factor $e^4 = (4\pi\alpha)^2 \approx 8.4 \times 10^{-3}$ to $|\mathcal{M}|^2$.

Phase space. Adding two massless particles to a k -body final state multiplies the phase-space volume by

$$\frac{\Phi_{k+2}}{\Phi_k} = \frac{m_\mu^4}{256 \pi^4 \cdot k(k+1)(k-1)(k-2)!/(k-2)!}, \quad (2)$$

which ranges from 6.9×10^{-11} ($k = 3 \rightarrow 5$) to 2.1×10^{-12} ($k = 7 \rightarrow 9$). The phase-space penalty increases with multiplicity because the available energy must be shared among more particles.

Diagram multiplicity. The number of dominant-class diagrams grows as $1 \rightarrow 4 \rightarrow 84 \rightarrow 4,320$ (ratios: 4, 21, 51), partially compensating the coupling and phase-space suppression. Each new pair can attach its photon to any of the existing charged lines, and the identical-particle permutations multiply rapidly.

The net per-pair suppression factors are:

Table 4: Per-pair suppression factor $\text{BR}(n+1)/\text{BR}(n)$ from NDA, MadGraph, and analytical estimates.

Transition	NDA	MadGraph	$(\alpha/\pi)^2$
$n=0 \rightarrow 1$	6.5×10^{-6}	3.6×10^{-5}	5.4×10^{-6}
$n=1 \rightarrow 2$	1.0×10^{-4}	1.2×10^{-5}	5.4×10^{-6}
$n=2 \rightarrow 3$	4.4×10^{-4}	—	5.4×10^{-6}

The NDA per-pair suppression weakens with each step (from 6.5×10^{-6} to 4.4×10^{-4}) because the diagram multiplicity growth outpaces the additional phase-space penalty. The MadGraph results show a more stable suppression $\sim 10^{-5}$ per pair, closer to the naive $(\alpha/\pi)^2 \approx 5.4 \times 10^{-6}$ estimate. This suggests that destructive interference among the proliferating diagrams at higher n partially cancels the combinatorial enhancement that NDA assigns by summing diagram contributions incoherently.

Figure 4 decomposes the per-pair suppression into its constituent factors. The coupling factor e^4 is constant; the phase-space ratio decreases (more suppressive) with n ; and the diagram

multiplicity ratio increases (less suppressive). The near-cancellation of these competing trends produces the remarkably stable $\sim 10^{-5}$ effective suppression per pair seen in the MadGraph results.

5 Experimental frontier

Table 5 synthesizes all results and identifies the observability status at each multiplicity.

Table 5: Comprehensive summary of multi-pair muon decay. Branching ratios marked with † are MadGraph exact values; others are NDA estimates. “Expected events” assumes the full Mu3e Phase I dataset (2.5×10^{15} muons).

n	Final state	Total diags	Dominant ($1W$)	BR	Expected events	Status
0	$e^+\nu\bar{\nu}$	1	1	~ 1	2.5×10^{15}	Measured
1	$e^+e^+e^-\nu\bar{\nu}$	18	4	$3.56 \times 10^{-5\dagger}$	8.9×10^{10}	Measured
2	$3e^+2e^-\nu\bar{\nu}$	1,122	84	$4.34 \times 10^{-10\dagger}$	1.1×10^6	Mu3e Phase I
3	$4e^+3e^-\nu\bar{\nu}$	149,400	4,320	$\sim 10^{-13}$	~ 250	Mu3e Phase I
4	$5e^+4e^-\nu\bar{\nu}$	$\sim 10^7$	$\sim 3 \times 10^5$	$\sim 10^{-16}$	~ 0.3	Phase II / HiMB

$n = 2$ (**first observation**). With $\text{BR} = 4.34 \times 10^{-10}$ and 2.5×10^{15} muon stops at Mu3e Phase I, approximately 10^6 signal events are expected — a precision measurement, not just a discovery. The five charged tracks in the final state provide a distinctive experimental signature.

$n = 3$ (**frontier**). The NDA estimate gives $\text{BR} \sim 8 \times 10^{-13}$, which based on the NDA/MG calibration at lower multiplicities likely corresponds to a true branching ratio in the range 10^{-14} – 10^{-12} . Even at the conservative end, this yields ~ 25 – $2,500$ events at Mu3e Phase I, making $n = 3$ observable with seven charged tracks. The event yield grows by a factor of ~ 20 at Phase II.

$n = 4$ (**beyond the frontier**). Extrapolating the per-pair suppression, we estimate $\text{BR}(n=4) \sim 10^{-17}$ – 10^{-16} . The lower end lies below Mu3e Phase II sensitivity ($\text{SES} \sim 10^{-16}$) but could potentially be accessible with the full HiMB upgrade ($\text{SES} \sim 10^{-18}$). The nine charged tracks would provide an unmistakable signature.

$n \geq 5$. At $\text{BR} \lesssim 10^{-20}$, the $n = 5$ channel is far below any foreseeable muon experiment. We identify $n = 4$ as the absolute upper limit of observability.

6 Discussion

6.1 Physics of the multiplicity scaling

The multi-pair branching ratio obeys an approximate scaling law

$$\text{BR}(n) \sim \left(\frac{\alpha}{\pi}\right)^{2n} \times C(n), \quad (3)$$

where $C(n)$ is a combinatorial enhancement factor that grows with n due to the proliferating Feynman diagram topologies. The key physical insight is that each additional e^+e^- pair is

produced by a virtual photon with momentum $q \lesssim m_\mu/2$ converting via $\gamma^* \rightarrow e^+e^-$. The conversion probability per photon is approximately

$$P_{\text{conv}} \sim \frac{2\alpha}{3\pi} \ln\left(\frac{m_\mu}{4m_e}\right) \approx 0.019, \quad (4)$$

comparable to the Kroll–Wada result for internal conversion. However, the actual per-pair suppression in the branching ratio is smaller ($\sim 10^{-5}$) because it also includes the phase-space penalty for distributing the muon’s rest-mass energy among additional particles.

6.2 Diagram proliferation and interference

The contrast between NDA and MadGraph illuminates the role of quantum interference. NDA sums diagram contributions incoherently (adding widths), while MadGraph computes the full squared amplitude with all interference terms. At $n = 1$, NDA underestimates by a factor of 2, suggesting constructive interference among the 4 dominant-class diagrams. At $n = 2$, NDA overestimates by a factor of 4, indicating that destructive interference among the 84 dominant-class diagrams reduces the rate below the incoherent sum.

This transition from constructive to destructive interference as n increases has a natural interpretation: at low multiplicity, the few diagrams share similar kinematic configurations and interfere constructively; at high multiplicity, the many diagrams sample a wider variety of kinematic configurations, and the random-sign phases produce partial cancellation. This effect partially counteracts the diagram multiplicity growth, stabilizing the effective per-pair suppression near 10^{-5} .

6.3 Implications for Mu3e

The SM multi-pair decays constitute an irreducible background for Mu3e’s primary physics goal (lepton-flavor-violating $\mu \rightarrow eee$) at $n = 1$, and they provide calibration signals at higher n . The $n = 2$ channel in particular offers a distinctive five-track signature with a known branching ratio, ideal for validating tracking efficiency and vertex reconstruction. The observation of $n = 3$ would represent the first measurement of a nine-body weak decay — a milestone in multi-body final-state physics.

7 Conclusions

We have systematically enumerated Feynman diagrams, estimated branching ratios via NDA, and cross-checked with MadGraph for the SM decay $\mu^+ \rightarrow \bar{\nu}_\mu \nu_e e^+ + n(e^+e^-)$ at pair multiplicities $n = 0\text{--}3$, with extrapolations to $n = 4$.

The experimental frontier for multi-pair muon decay is at $n = 3$ (seven charged tracks), which should be observable at Mu3e Phase I with $\sim 10^2\text{--}10^3$ events. The $n = 4$ channel may be marginally accessible at Mu3e Phase II or the HiMB upgrade. Channels with $n \geq 5$ lie beyond the reach of any planned experiment.

The per-pair suppression factor of $\sim 10^{-5}$ results from the interplay of the QED coupling ($e^4 \approx 8 \times 10^{-3}$), phase-space compression ($\Phi_{k+2}/\Phi_k \sim 10^{-11}\text{--}10^{-12}$), and diagram multiplicity growth (factors of 4–51), with quantum interference effects modulating the net suppression to a remarkably stable value across multiplicities.

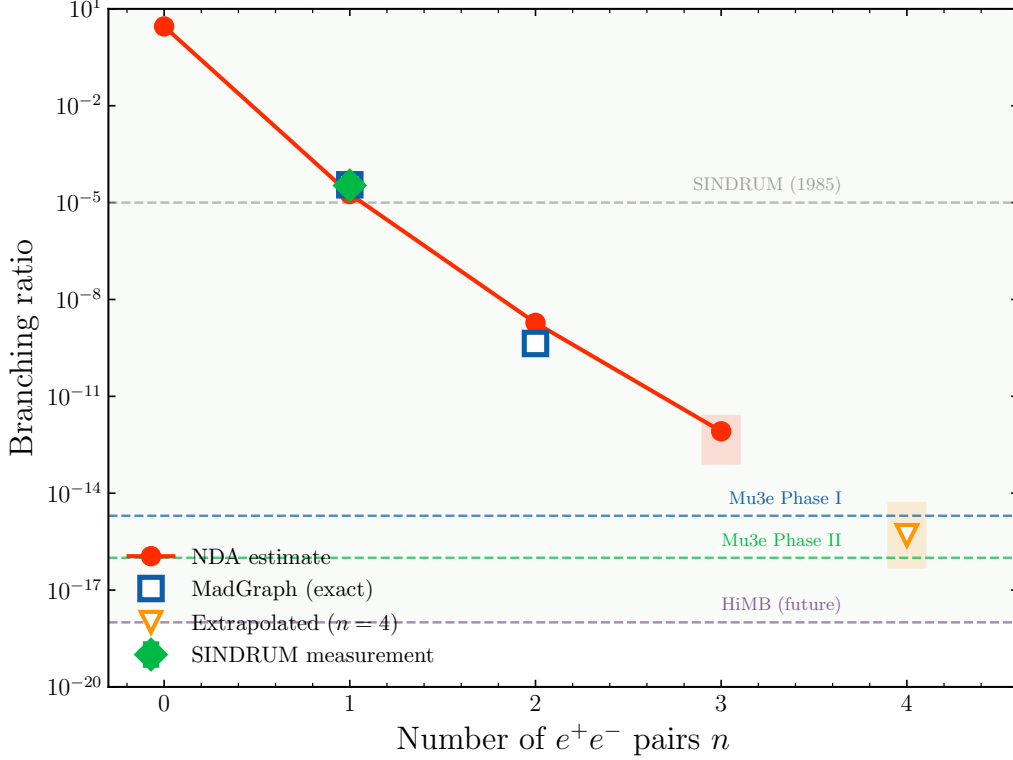


Figure 2: Branching ratio of $\mu^+ \rightarrow \bar{\nu}_\mu \nu_e e^+ + n(e^+ e^-)$ as a function of the number of electron-positron pairs n . NDA estimates (red circles), MadGraph exact results (blue squares), and the SINDRUM measurement (green diamond) are shown alongside experimental sensitivity thresholds for SINDRUM, Mu3e Phase I and II, and HiMB. The $n = 4$ point (orange triangle) is extrapolated from the per-pair suppression factor. Shaded bands indicate NDA uncertainty.

Figures

References

- [1] SINDRUM Collaboration, “Search for the decay $\mu^+ \rightarrow e^+ e^+ e^-$,” Nucl. Phys. B **299** (1988) 1.
- [2] G. M. Pruna, A. Signer, and Y. Ulrich, “Fully differential NLO predictions for the radiative decay of muons and taus,” Phys. Lett. B **772** (2017) 452, arXiv:1611.03726.
- [3] M. Fael and M. Passera, “Muon–electron scattering at NLO,” Phys. Rev. Lett. **122** (2019) 192001, arXiv:1811.10965.
- [4] M. Hostert, T. Menzo, M. Pospelov, and J. Zupan, “New physics in multi-charged-lepton final states of muon and tau decays,” JHEP **2023** 129, arXiv:2306.15631.
- [5] Mu3e Collaboration, “Technical design report for the Mu3e experiment,” arXiv:2009.11690 (2020).
- [6] Mu3e Collaboration, “Status of the Mu3e experiment,” arXiv:2501.14667 (2025).
- [7] A. Aiba *et al.*, “Science case for the new High-Intensity Muon Beams HIMB at PSI,” arXiv:2111.05788 (2021).

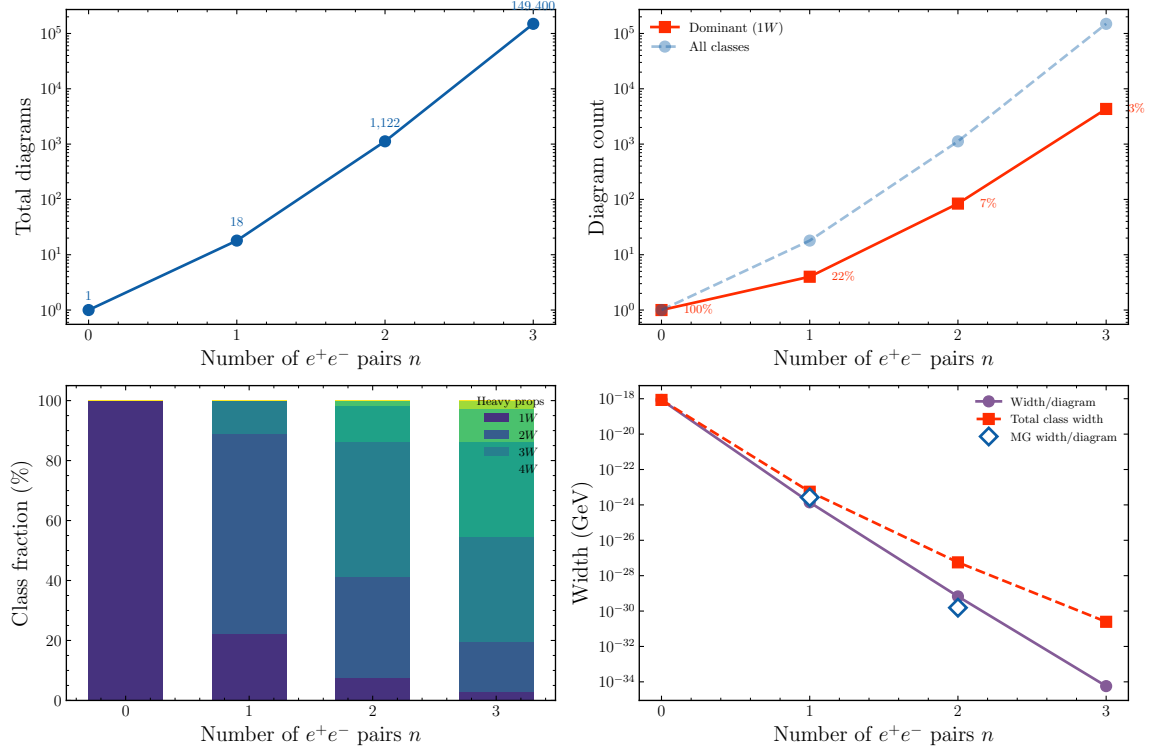


Figure 3: Diagram enumeration metadata. Upper left: total Feynman diagram count vs. pair multiplicity. Upper right: dominant-class (single- W) diagram count compared to the total, with percentages. Lower left: fractional composition by heavy-propagator class, showing the proliferation of multi- W topologies that are nonetheless negligible in the rate. Lower right: NDA width per diagram (purple) and total dominant-class width (red dashed) compared to MadGraph per-diagram widths (blue diamonds).

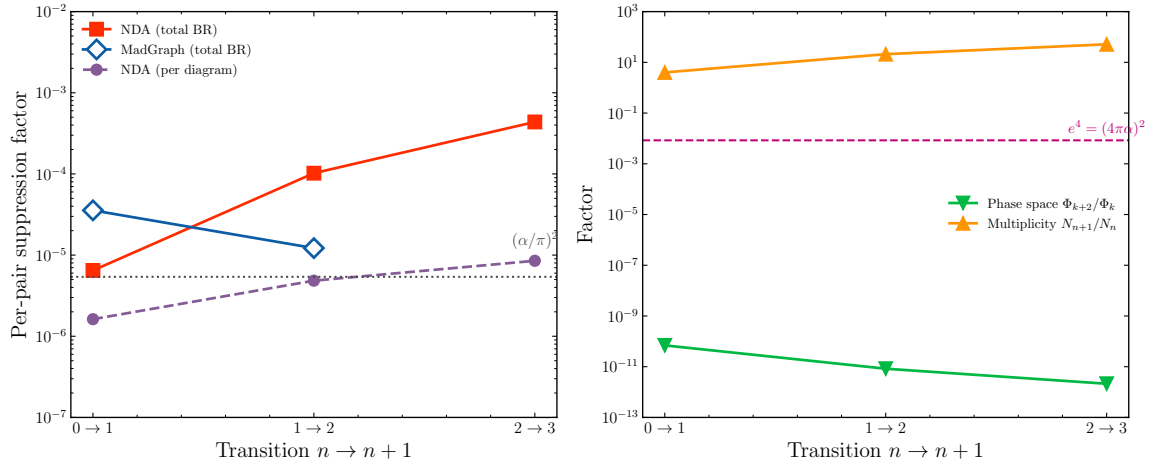


Figure 4: Per-pair suppression analysis. Left: effective branching ratio suppression per additional e^+e^- pair from NDA (red squares), MadGraph (blue diamonds), and NDA per-diagram (purple circles), compared to the $(\alpha/\pi)^2$ reference (dotted). Right: decomposition into coupling (e^4 , constant), phase-space ratio (Φ_{k+2}/Φ_k , decreasing), and diagram multiplicity ratio (N_{n+1}/N_n , increasing). The near-cancellation of phase space and multiplicity trends produces the stable effective suppression.